Student Name (in English): _____ Student ID: _____

PLO K1-CLO 1.1
Question 1: MCQs To be answered in (20) Minutes [ ] / 10 Marks

Choose the best answer (if any) to complete the meaning of the following statements. **FILL THE TABLE BELOW, ONLY ONE CHOICE IS ACCEPTED.**

1	2	3	4	5	6	7	8	9	10

1. **The following is an example of an $O(1)$ time complexity algorithm:**
   A. Traversing all elements of a linked list.
   B. Accessing the top element of a stack.
   C. Inserting an element into a sorted array, while maintaining the sorted order.
   D. Reversing a string.

2. **Which of the following expressions grows the fastest as $n \to \infty$ (worst efficiency):**
   A. $n\ log(n)$
   B. $\sqrt{n}\ log(n)$
   C. $n\ log^2(n)$
   D. $log^2(n) + nlog\ (n)$

3. **Given the undirected graph with vertices {A, B, C, D, E} and weighted edges:**
   A–B (2), A–C (2), B–C (3), B–D (1), C–D (1), D–E (4), C–E (5)
   If Prim's algorithm starts at vertex A, and ties are broken alphabetically, which node is the third to be added to the growing MST?
   A. Node C
   B. Node D
   C. Node B
   D. Node E

4. **When using Counting Sort for a dataset with range 0 to 10^6 and only 100 elements, what is the biggest issue?**
   A. Time complexity
   B. Space complexity
   C. Instability
   D. It's comparison-based

5. **In Radix Sort, how is Counting Sort typically applied?**
   A. Applied once after all digit passes to finalize the sorted order.
   B. Applied before each digit pass to arrange elements by future digits.
   C. Applied at each digit from least to most significant to ensure stability.
   D. Applied at each digit from most to least significant to ensure stability

6. **What is the time complexity of Dijkstra's algorithm using a min-heap priority queue?**
   A. $O(|V|^2\ lg\ |E|))$
   B. $O(|E|)$
   C. $O((|V|\ + |E|)\ lg\ |V|)$
   D. $O(|V| lg\ (|E| + |V|))$

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7. **In Bellman-Ford, how many edge relaxations are needed in the worst case?**
   A. $|V| - 1$ times over all E edges
   B. $|E| - 1$ times over all E edges
   C. $|V \lg V|$ times over all E edges
   D. $|V + E|$ times over all E edges

8. **Which principle allows dynamic programming to break problems into reusable subproblems?**
   A. Memorization
   B. Greedy
   C. Overlapping subproblems
   D. Divide and Conquer

9. ***If W represents* the knapsack's capacity and is larger than the total weight of all items, what does the *fractional* knapsack algorithm do?**
   A. *Takes no items*
   B. *Takes all items fully*
   C. *Takes only highest value item*
   D. *Fails*

10. ***Given the fractional* knapsack's capacity W = 30, and items with values and weights:**
   - **Item A: value = 60, weight = 10**
   - **Item B: value = 100, weight = 20**
   - **Item C: value = 120, weight = 30**

   ***Which item will be picked* partially in the optimal solution?**
   A. *Item A*
   B. *Item B*
   C. *Item C*
   D. *None*